Mayar (May) **Ahmed**

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Education

Department of Biological Sciences, University of Pittsburgh — *Pittsburgh, PA, USA*

Ph.D. Candidate in Molecular, Cellular, and Developmental Biology — *Aug 2021 – Expected Apr 2027*

- Advisor: **Dr. Jacob D. Durrant**
- Research focus: computational structural biology, cheminformatics, and machine learning for drug discovery.

University of Science and Technology, Zewail City — *Giza, Egypt*

B.S. in Computational Biology and Genomics (Biomedical Science) — *Aug 2017 – Jul 2021*

- Advisor: **Dr. Eman Badr**
- Research focus: computational genomics, bioinformatics, and machine learning for drug-response prediction.
- Thesis: *Investigating the correlation between molecular variations within acute myeloid leukemia (AML) patients and drug sensitivity.*

Research Experience

Graduate Student Researcher — *Pittsburgh, PA, USA*

Department of Biological Sciences, University of Pittsburgh — *Sep 2021 – Present*

- **LIGNOVA:** Designed and built an automated, open-source Python/SLURM pipeline that pairs PubChem bioactivity data with experimentally determined receptor structures and docks them with GNINA on HPC, producing **258M+ docked poses** (243,805 ligands, 11,268 protein structures, 5.86M protein–ligand pairs). 89% of pairs carry an experimental affinity label, and each pair is graded on pose reliability and affinity confidence into Gold/Silver/Bronze tiers for AI/ML training.
- **Explicit water vs. consensus rescoring in docking:** Designed and ran a benchmark on curated, high-quality X-ray complexes to test whether ligand **consensus rescoring** (ComBind) can replace explicit-water modeling in Glide pose prediction, comparing pose accuracy across three hydration conditions. **Manuscript in preparation.**
- **Structure-based discovery of arginine sensors in yeast:** (collaboration with the O'Donnell Lab) Led a structure-based reverse screen for arginine-sensing proteins in *S. cerevisiae* that sequence homology misses, combining Foldseek structural search against human CASTOR1 structures with MolModa (web-based AutoDock Vina) docking to prioritize **4 candidates**. Experimental follow-up by the O'Donnell Lab confirmed altered arginine-homeostasis phenotypes in a subset of candidates.
- **Comparative genomics of *Dictyostelium***: (collaboration with the Levin Lab) Led the computational genomics for a **co-first-author *PNAS* study**, debugging and adapting an assembly pipeline for HPC and developing the annotation, quality control, and targeted *tgrB1*/*tgrC1* locus analysis across 10 chromosome-length genomes.

Undergraduate Student Researcher — *Giza, Egypt*

University of Science and Technology, Zewail City — *Jun 2020 – Apr 2021*

- Trained Random Forest regression models (scikit-learn) to predict **drug sensitivity from transcriptomic profiles** in acute myeloid leukemia patients, identifying genetic markers correlated with differential drug response.

Software

LIGNOVA	Lead developer. Fully automated, open-source pipeline that handles all receptor and ligand preparation, then docks bioactive compounds against experimentally determined protein structures to build quality-graded ML data
reqadence	Lead developer. Reusable async base class for REST API clients with retries, rate limiting, and RFC 9111 caching; ships a built-in RCSB PDB client and is subclassed by LIGNOVA's PubChem client for large-scale data ingestion
dicty_genomes	Contributor. Adapted the assembly pipeline for SLURM; wrote annotation and QC modules (tandem repeats, telomeres, TEs, rRNA, mtDNA, variant calling, CRAQ) and the *tgrBC* locus analysis behind the *PNAS* 2025 study

Publications

Holland, M.*, **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, *122*(51). DOI: 10.1073/pnas.2520843122. (*co-first authors)

Kochnev, Y.; **Ahmed, M.**; Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, *52*(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, *21*(1), 299. DOI: 10.1186/s12915-023-01791-z.

Honors & Awards

2026	**NSF ACCESS Allocation (BIO260240)**, Purdue Anvil & TAMU ACES compute to scale LIGNOVA	*USA*
2026	**GPSG Travel Grant**, Graduate & Professional Student Government, University of Pittsburgh	*Pittsburgh, PA*
2026	**Elizabeth Baranger Excellence in Teaching Award Nominee**, Biochemistry; nominated 2024 & 2026	*Pittsburgh, PA*
2021	**Provost's Honors Roll**, Top 5 students by cumulative GPA, Zewail City	*Giza, Egypt*
2021	**Cum Laude**, B.S. Computational Biology & Genomics, Zewail City	*Giza, Egypt*
2017–2021	**Merit Scholarship (90%)**, Four-year undergraduate merit scholarship, Zewail City	*Giza, Egypt*

Conferences & Talks

2026	**Poster Presenter**: LIGNOVA: An Automated Pipeline for Large-Scale Generation of High-Quality Protein–Ligand Complexes, Computational Medicinal Chemistry School	*Cambridge, MA*
2026	**Speaker**: Putting the Pieces Together *in silico* to Build Drugs from Fragments, MCDB Noon Seminar Series, University of Pittsburgh	*Pittsburgh, PA*
2024	**Speaker**: DeepFrag Meets LIGNOVA: Transforming Drug Design with Fragment-Based Lead Optimization, MCDB Noon Seminar Series, University of Pittsburgh	*Pittsburgh, PA*
2024	**Poster Presenter**: Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, Gordon Research Conference & Seminar on Computational Chemistry	*Portland, ME*
2024	**Speaker**: Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, MCDB Noon Seminar Series, University of Pittsburgh	*Pittsburgh, PA*
2023	**Poster Presenter**: Augmenting Protein–Ligand Complex Databases with PubChem for Enhanced Drug Discovery, Gordon Research Conference & Seminar on Computer-Aided Drug Design	*West Dover, VT*
2022	**Speaker**: Testing the Impact of a Newly Built Receptor–Ligand Dataset on DeepFrag Accuracy, MCDB Noon Seminar Series, University of Pittsburgh	*Pittsburgh, PA*

Skills

Programming	Python, Bash, R, MATLAB; object-oriented design
Software Engineering	Git/GitHub, GitHub Actions, pytest, Ruff, Black, pixi, MkDocs; Linux, SLURM, HPC
Machine Learning	Supervised learning, feature engineering, model evaluation, graph neural networks, Random Forest
Packages	PyTorch, scikit-learn, NumPy, pandas, Polars, PyArrow, SciPy, RDKit, Biopython, MDAnalysis, httpx
Data Analysis	Statistical modeling; data visualization (Matplotlib, Seaborn); Parquet
Cheminformatics	Docking (Glide, GNINA, AutoDock Vina, MolModa), rescoring (ComBind), structural search (Foldseek), structure prediction (AlphaFold2), preparation (PDB2PQR, Gypsum-DL, Meeko, Open Babel), visualization (UCSF ChimeraX)
Genomics	Assembly (Flye; Nanopore/Illumina), alignment (minimap2, BWA, samtools), variant calling (bcftools, bedtools), annotation (TRF, RepeatMasker, BLAST), QC (BUSCO, CRAQ, FastQC), synteny (MCScanX)
Languages	English (fluent), Arabic (native)

Teaching & Mentorship Experience

Undergraduate Research Mentor — *Pittsburgh, PA, USA*

TECBIO REU, Department of Computational & Systems Biology, University of Pittsburgh — *May 2024 – Jul 2024*

- Co-mentored an undergraduate student on a 10-week project applying AlphaFold2 to explore HIV protease conformations for drug design.
- Supervised development of a research poster, presented at the Research Symposium at Duquesne University.

Graduate Teaching Assistant & Fellow (Biochemistry) — *Pittsburgh, PA, USA*

Department of Biological Sciences, University of Pittsburgh — *Spring 2024, Fall 2025*

- Conducted recitation sections and held weekly office hours for ~200 students; mentored 15 undergraduate teaching assistants.
- Nominated by undergraduate students for the **Elizabeth Baranger Excellence in Teaching Award** (2024, 2026).

Undergraduate Teaching Assistant (Python) — *Giza, Egypt*

University of Science and Technology, Zewail City — *Sep 2020 – Jan 2021*

- Guided students in writing and debugging Python code; held weekly office hours.

Leadership & Service

The Dietrich School of Arts & Sciences Graduate Student Organization — *Pittsburgh, PA, USA*

Department of Biological Sciences Representative — *Aug 2024 – Sep 2026*

- Represented Biological Sciences at school-wide GSO meetings, addressing department concerns.
- Coordinated with the GSO executive board on event planning, policy discussions, and decision-making processes.
- **Served on the Summer Research Grant panel**, evaluating graduate student research proposals across different A&S departments.

Pitt Biological Sciences Graduate Student Organization *Pittsburgh, PA, USA*

Graduate Diversity, Inclusion, and Belonging (DIB) Committee Member *Sep 2023 – Dec 2024*

- DIB works on fostering an inclusive environment for underrepresented students in the department.
- Organized intercultural celebrations and distributed informational resources highlighting the department's diverse cultural traditions.

Zewail City OpenCourseWare *Giza, Egypt*

Vice President *Jan 2018 – Mar 2020*

- Led a team making university course materials (lecture notes, videos, assignments) freely available online for Egyptian students.
- Selected courses to document each semester, coordinated sub-teams, and contributed LaTeX-typeset lecture notes and solutions.

Bio Team Egypt *Egypt*

Head of Strategy Committee *Jan 2017 – Sep 2017*

- Managed 3 subcommittees (Events, Awareness Campaigns, Charities) and organized 3 biotechnology seminars.
- Coordinated 4 public awareness campaigns introducing biotechnology to high school students across Egypt.